

CS2311 Modern Operating System Exercises Spring 2026

Exercise 2: Process Synchronization

Due date: April 9, 2026

1. You are designing a data structure for efficient dictionary lookup in a multithreaded application. The design uses a hash table that consists of an array of pointers each corresponding to a hash bin. The array has 1001 elements, and a hash function takes an item to be searched and computes an entry between 0 and 1000. The pointer at the computed entry is either null, in which case the item is not found, or it points to a doubly linked list of items that you would search sequentially to see if any of them matches the item you are searching for. There are three functions defined on the hash table: Insertion (if an item is not there already), Lookup (to see if an item is there), and deletion (to remove an item from the table). Considering the need for synchronization, would you:
 - a. Use a mutex over the entire table?
 - b. Use a mutex over each hash bin?
 - c. Use a mutex over each hash bin and a mutex over each element in the doubly linked list?

Please justify your answer.

2. SJTU student applicants stand in a line in the Admissions office to pick application forms. Initially, there is a pile of 50 forms on a desk. Once one is the first in the line, an applicant checks the desk. If there are application forms, she pick one and leave. If no application form is left on the desk, the applicant rings a bell. An Admissions officer hears the bell and places a new pile of 50 forms on the desk.

Questions:

- a. Declare all the variables/arrays you need for this problem and initialize them.
 - b. Declare all the semaphores needed for synchronization and mutual exclusion you need for this problem and initialize them.
 - c. Write two procedures: one indicates what an applicant does, and the other indicates what the Admissions officer does.
3. A club has a lounge where the members can sit and chat. Members include both smokers and non-smokers. Smokers can smoke in the lounge when non-smokers are absent. Device a protocol for the lounge. A smoker calls enter Lounge (true) to enter the lounge (the flag true indicates that she is a smoker), then calls smoke(), and finally calls leave Lounge(true) to leave the lounge. Similarly, a non-smoker calls enter Lounge (false) to enter, then sits in the lounge and chats with others, and finally calls leave Lounge (false) to leave the lounge.

Questions:

- a. Declare all the variables/arrays you need for this problem and initialize them.
 - b. Declare all the semaphores needed for synchronization and mutual exclusion you need for this problem and initialize them.
 - c. Write pseudo code for the functions noted above so that smoking rules are obeyed.
4. Search-Insert-Delete Problem

Three kinds of processes share access to a data structure: searchers, inserter, and deleters. Searchers access the list without changing it. So, any number of concurrent searchers can be accessing the structure safely. Inserters have the ability to add new elements to the end of the structure. Only one inserter can access the structure at any given time, but can work concurrently with any number of searchers. Lastly, Deleters can remove items from any position in the structure. As such, any deleter demands exclusive access to the structure. So, while a deleter has access to the structure neither any other deleter, any inserter, nor any searcher is allowed access.

- a. Declare all the variables/arrays you need for this problem and initialize them.
- b. Declare all the semaphores needed for synchronization and mutual exclusion you need for this problem and initialize them.
- c. Write the pseudo code for this problem to prevent starvation.